1. Hyperploidy

Addition of one or more chromosomes to diploid sets are called **hyperploidy**. Diploid set of chromosomes represented as Disomy. Hyperploidy can be divided into three types. They are as follows,

(a) Trisomy

Addition of single chromosome to diploid set is called **Simple trisomy**(2n+1). Trisomics were first reported by Blackeslee (1910) in ***Datura stramonium*** (Jimson weed). But later it was reported in ***Nicotiana, Pisum*** and ***Oenothera.*** Sometimes addition of two individual chromosome from different chromosomal pairs to normal diploid sets are called **Double trisomy** (2n+1+1).

(b) Tetrasomy

Addition of a pair or two individual pairs of chromosomes to diploid set is called **tetrasomy** (2n+2) and **Double tetrasomy** (2n+2+2) respectively. All possible tetrasomics are available in Wheat.

(c) Pentasomy

Addition of three individual chromosome from different chromosomal pairs to normal diploid set are called pentasomy (2n+3).

2. Hypoploidy

Loss of one or more chromosome from the diploid set in the cell is called **hypoploidy**. It can be divided into two types. They are

(a) Monosomy

Loss of a single chromosome from the diploid set are called **monosomy**(2n-1). However loss of two individual or three individual chromosomes are called **double monosomy** (2n-1-1) and **triple monosomy** (2n-1-1-1) respectively. Double monosomics are observed in maize.

(b) Nullisomy

Loss of a pair of homologous chromosomes or two pairs of homologous chromosomes from the diploid set are called **Nullisomy** (2n-2) and **double Nullisomy** (2n-2-2) respectively. Selfing of monosomic plants produce nullisomics. They are usually lethal.

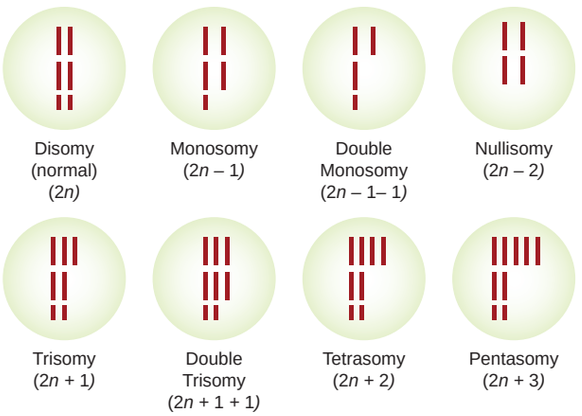


Figure 3.21 Types of aneuploidy

(ii) Euploidy

Euploidy is a condition where the organisms possess one or more basic sets of chromosomes. Euploidy is classified as monoploidy, diploidy and polyploidy. The condition where an organism or somatic cell has two sets of chromosomes are called diploid (2n). Half the number of somatic chromosomes is referred as gametic chromosome number called haploid(n). It should be noted that haploidy (n) is different from a monoploidy (x). For example, the common wheat plant is a polyploidy (hexaploidy) 2n=6x=72 chromosomes. Its haploid number (n) is 36, but its monoploidy (x) is 12. Therefore, the haploid and diploid condition came regularly one after another and the same number of chromosomes is maintained from generation to generation, but monoploidy condition occurs when an organism is under polyploidy condition. In a true diploid both the monoploid and haploid chromosome number are same. Thus a monoploid can be a haploid but all haploids cannot be a monoploid.

Polyploidy

Polyploidy is the condition where an organism possesses more than two basic sets of

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